Electrical Power Systems

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Electrical Power Systems (A13089)

Short Title:	Electrical Power Systems		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electrical Engineering		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

The aim of this module is to provide the student with a professional level of understanding and the ability to apply relative design and calculation methods to the areas of electrical power generation, transmission, and distribution equipment. In addition, the student will gain proficiency in the safety and maintenance requirements for electrical medium and high voltage electrical systems taught.

Programmes

`ELEC-0041`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	3 / 5 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	3 / 6 / M
`ELEC-0041`	BEng (Hons) (`WD_ETRIC_B`)	3 / 5 / M

Indicative Content

- Power generation methods, equipment control, protection and efficiency
- Power transmission / distribution topologies, equipment design, utilisation and efficiency calculations
- Networked power system control, status and metering
- Protective relays, strategies and associated transducers
- Preventative Maintenance and electrical safety
- Load, BIL, Power flow, Fault and stability calculations
- Sources and suppression of harmonic content.
- Typical projects: adaptive power factor correction, dynamic generator excitation, automated load-sharing systems, solid-state AC generation, black start strategic planning

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate and understand the design and operation of electrical transmission / distribution peripherals.
2. Determine transformer configuration for optimal energy conversion and management.
3. Evaluate and compare transmission and distribution topologies.
4. Format and compile preventative maintenance procedures and the safety requirements for typical medium voltage and high voltage equipment.
5. Analyse the results of typical distribution and transmission calculations, circuit design and simulation e.g. Load, BIL, Power flow, Fault and stability calculations.
6. Calculate generator rotor losses due to negative sequence generation.
7. Demonstrate the professional, ethical and effective communication skills of a professional engineer.

Learning and Teaching Methods

- The course will consist of lectures, laboratories, mini project and formative assessments

- The emphasis on course delivery will be to engage the student in the learning of the material, through the integration of the course materials, with the real case studies in laboratories and mini project
- The formative assignments will be used to provide an opportunity for the student to engage with the material, before it is applied, to maximise the learning opportunities for the student.
- Self-directed learning, moodle, and recommended software.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,5,6,7
Continuous Assessment	30%	
Lab Report	20%	3,4,5,6
Project	10%	5,7

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Grigsby, L.L. _Electric Power Generation, Transmission and Distribution_. 3rd. .: CRC Press, 2012.

Supplementary Material

- Bayliss, C. and B. Hardy. _Transmission and Distribution Electrical Engineering_. 4th Edition.. .: Elsevier, 2011.
- Bird, J.O. and A.J.C. May. _Standard Handbook for Electrical Engineers_. .: Elsevier, 1994.
- Warne, D.F. _Electrical Power Engineering Handbook_. 2nd Edition.. Oxford: Newnes, 2005.